Thin geometries from involution triples

Human Compression project: working application

We recover the Suzuki calculation in Section 5 of Jacobs and Leemans, *An algorithmic analysis of the intersection property*, LMS Journal of Computation and Mathematics 7 (2004), 284–299. Its input consists of 147 thin, residually connected, residually weakly primitive geometries for $G = \mathrm{Sz}(8)$. We obtain representatives without importing that list, and then apply the common shadow criterion. The output is 27 geometries with the intersection property, 48 with the weak property only, and 72 with neither. These are previously published results.

Throughout, the group preserves types and acts transitively on flags of each type. Incidence is symmetric and reflexive, with no incident distinct objects of one type. Every flag extends to a chamber. A geometry is thin if every flag missing one type has exactly two extensions. Residual connectedness requires the incidence graph of every residue of rank at least two to be connected. Residual weak primitivity means that in every nonempty residue at least one of its actions on a type is primitive.

Lemma 1. *Let a nonabelian finite simple group G act as above on a thin, residually connected geometry of rank three. For a chamber $C = \{p_0, p_1, p_2\}$, write $H_i = G_{p_i}$ and $B = \bigcap_i H_i$. There are distinct involutions t_0, t_1, t_2 such that*

$$B = 1, \qquad G = \langle t_0, t_1, t_2 \rangle, \qquad H_i = \langle t_j : j \neq i \rangle, \qquad H_i \cap H_j = \langle t_k \rangle \quad (\{i, j, k\} = \{0, 1, 2\}).$$

Proof. Put $R_i = \bigcap_{j \neq i} H_j$. Flag transitivity on the two extensions of $C \setminus \{p_i\}$ gives $[R_i : B] = 2$, so B is normal in each R_i.

The chamber graph is connected. To see this, a rank-two residue has a connected incidence graph, whose edges are its chambers; successive edges sharing a vertex give a gallery. In rank three take a path of incident objects between a vertex of one chamber and a vertex of another. Extend each edge to a chamber. Consecutive chosen chambers contain a common object, and the rank-two residue of that object supplies a gallery between them. The end chambers connect in the same way. Thus all chambers are connected by changes of one object.

The chambers adjacent to C across its i-panel are images of C under R_i. Transporting along a gallery proves $G = \langle R_0, R_1, R_2 \rangle$: the generated subgroup reaches every chamber, and it contains their common base stabilizer B. Therefore $B \lhd G$. Since $B < R_i$, simplicity gives $B = 1$. Each R_i now has a unique nonidentity element t_i, an involution. Within the residue of p_i, the same gallery argument gives $H_i = \langle R_j : j \neq i \rangle$. Distinctness follows because an element in two R_i lies in B. The stated intersections follow from the stabilizer definitions. □

Lemma 2. *Conversely take three distinct involutions generating G, and put $H_i = \langle t_j : j \neq i \rangle$. Suppose the three intersections in Lemma 1 hold. Their coset incidence system is a flag-transitive geometry if and only if, for each ordering i, j, k of the types,*

$$(H_i H_j) \cap (H_i H_k) = H_i (H_j \cap H_k). \tag{1}$$

It then is thin and residually connected. It is residually weakly primitive if and only if at least one H_i is maximal in G and all three numbers $|H_i|/2$ are prime.

Proof. Any incident pair of cosets has a common element, so G is transitive on incident pairs. Normalize the cosets of types j, k to H_j, H_k. The cosets $H_i g$ incident with both correspond

exactly to the elements of $(H_iH_j) \cap (H_iH_k)$. They extend to chambers in the base chamber's orbit precisely when they meet $H_j \cap H_k$. This gives (1). Every flag then extends to such a chamber. The right side is contained in the left side; since the triple intersection is trivial, its size is $2|H_i|$. Thus equality is checked without constructing an entire incidence graph.

The base chamber stabilizer is trivial and its panel stabilizers have order two, so the geometry is thin. Each H_i is generated by the two panel stabilizers within it; the whole group is generated by all three. Their coset incidence graphs are connected, proving residual connectedness. Each rank-two residue is the polygon associated to a dihedral group of order $2m_i$, where $m_i = |H_i|/2$. Its vertex stabilizers are reflections. A reflection subgroup is maximal in a dihedral group of order $2m_i$ exactly when m_i is prime: intermediate subgroups correspond to proper divisors of the rotation order. Rank-one residues have degree two and are primitive. The remaining condition is primitivity on some type of the whole geometry, equivalent to maximality of its stabilizer H_i. □

A finite enumeration with an explicit orbit reduction

The ambient permutation group is the previously validated degree-65 model used in the rank-two Suzuki calculation. We retain that model and GAP's exact finite group algorithms. The computation checks order 29120 and simplicity, one conjugacy class containing all 455 involutions, and an automorphism group of order 87360. Its action on those involutions has trivial kernel. It also computes representatives of all maximal subgroups, whose orders are $448, 52, 20, 14$. These are explicit computational premises; their implementations have not been formalized in Lean here.

Let $\mathcal{A}$ be the computed automorphism group acting on the 455 involutions, numbered in a fixed order. Fix the first involution. For its stabilizer $\mathcal{A}_1$, take one representative b from each orbit on the other involutions. For each b, take one c from each orbit of $\mathcal{A}_{1,b}$ on the remaining involutions. This gives all ordered triples up to $\mathcal{A}$, exactly once.

To forget the type order, normalize each of the six permutations of a triple by the same stabilizer procedure and retain the smallest resulting pair of indices (b, c). The normalization is independent of transporter choices: after placing the first entry at 1, any two choices differ by $\mathcal{A}_1$; after placing the second entry at b, they differ by $\mathcal{A}_{1,b}$. Thus the final orbit of the third entry is fixed. Equality of the minimum pairs is equivalent to equivalence under $\mathcal{A}$ and a permutation of the three types.

For each retained triple the program checks generation, the exact subgroup intersections, all three product equalities (1), and the primitivity conditions of Lemma 2. Top primitivity is checked by conjugacy to the computed maximal subgroups, not inferred from a subgroup order. The resulting counts are

Stage	Count
Ordered triples modulo $\mathcal{A}$	1083
After forgetting the three type names	195
Generating triples	186
Triples satisfying all marked subgroup intersections	186
Flag-transitive thin geometries	183
Residually weakly primitive geometries	147

The three rejected generating intersection systems show why a subgroup intersection test alone does not establish the required geometry.

For these thin connected geometries the type-preserving automorphism group is exactly G. Indeed an automorphism fixing one chamber fixes its unique neighbors across every panel, and then every chamber by connectedness. The full type-preserving group is consequently

semiregular on chambers, while G is already transitive. It can have no more elements than G. Thus an isomorphism between two such geometries induces an automorphism of G and a permutation of types, exactly the equivalence used above.

Shadow tests and their limits

For each retained triple, the common shadow theorem expresses every shadow as an orbit of a chamber stabilizer intersection in $H_i \backslash G$. It checks all normalized object–flag pairs and every allowable witness flag. The implementation is the earlier reviewed shadow implementation, with the ambient order and input triples changed. It caches conjugate coset actions: if $H' = H^g$, the base point changes from p to p^g. The action is unchanged, and the stabilizer of the new point is checked to equal H'. No shadow or witness is removed by this optimization.

The full eligible-extension test is repeated in all three coset actions for each geometry, providing 1764 flag-extension identities in addition to the product tests above. Each of the 441 type tests is recorded. The resulting numbers are 27 with all three types passing, 48 with some but not all types passing, and 72 with no type passing. All rank-two residues are polygons of lengths $2m_i$ with $m_i \in \{2, 5, 7, 13\}$. For $m_i > 2$ there is no quadrilateral: shadows of distinct opposite-type vertices meet in at most one object, which itself is a witness whenever present. If $m_i = 2$, the residue is the complete bipartite graph $K_{2,2}$; the two opposite-type shadows are the entire target type, with the empty flag as witness. Flags already containing the target type have singleton shadows and use that object as witness; the empty input flag has the whole type as shadow. This proves the rank-two intersection property in both cases. Rank-one residues are automatic. Consequently the rank-three answers are also the full residual answers.

The orbit and product arguments enumerate all *thin* geometries with the stated hypotheses. They do not prove that every residually weakly primitive rank-three geometry of Sz(8) is thin. That separate classification assertion is retained from the earlier classification cited by Jacobs and Leemans. We claim no recovery of its unavailable manuscript. The present result is the reported shadow split on its 147 thin inputs.

The thin chamber argument and stabilizer orbit enumeration are standard regular-geometry and permutation-group methods. The shadow normalization and geometric intersection properties are from Jacobs and Leemans. Our contribution here is an explicit common implementation and an independently generated list of inputs. Neither the count nor its mathematical priority is new. This draft awaits independent review; it adds no original-manuscript credit, numerical compression factor or complete Lean formalization.